10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 3.5 — Determinant, Adjoint & Inverse

Unit 3 | Solved Exercise | New Book (PECTAA)

Exercise 3.5 (Solved)

1. Find the values of each of the determinants.

For a matrix A = [[a, b], [c, d]] the determinant is |A| = ad − bc. Multiply the main diagonal and subtract the product of the other diagonal.

(i) $\begin{vmatrix} 10 & 5 \\ 4 & 6 \end{vmatrix}$

$|A| = (10)(6) - (5)(4)$

$= 60 - 20$

Answer: |A| = 40

(ii) $\begin{vmatrix} -5 & 8 \\ -3 & -7 \end{vmatrix}$

$|A| = (-5)(-7) - (8)(-3)$

$= 35 - (-24)$

$= 35 + 24$

Answer: |A| = 59

(iii) $\begin{vmatrix} 3 & 8 \\ 0 & 2 \end{vmatrix}$

$|A| = (3)(2) - (8)(0)$

$= 6 - 0$

Answer: |A| = 6

2. Find whether the following matrices are singular or non-singular.

A square matrix is singular if its determinant is zero, and non-singular if its determinant is not zero.

$A = \begin{bmatrix} 5 & 3 \\ 3 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 7 & 21 \\ 2 & 6 \end{bmatrix}$ $C = \begin{bmatrix} 13 & 5 \\ 7 & 3 \end{bmatrix}$ $D = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$

studysapiens.com

Matrix A

$$|A| = \begin{vmatrix} 5 & 3 \\ 3 & 2 \end{vmatrix}$$

$= (5)(2) - (3)(3)$

$= 10 - 9$

$= 1$

$|A| \neq 0$, so A is non-singular

Matrix B

$$|B| = \begin{vmatrix} 7 & 21 \\ 2 & 6 \end{vmatrix}$$

$= (7)(6) - (21)(2)$

$= 42 - 42$

$= 0$

$|B| = 0$, so B is singular

Matrix C

$$|C| = \begin{vmatrix} 13 & 5 \\ 7 & 3 \end{vmatrix}$$

$= (13)(3) - (5)(7)$

$= 39 - 35$

$= 4$

$|C| \neq 0$, so C is non-singular

Matrix D

$$|D| = \begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix}$$

$= (2)(0) - (0)(0)$

$= 0$

$|D| = 0$, so D is singular

Answer: Singular: B and D Non-singular: A and C

studysapiens.com

3. Find the value of x when A = [[x, 6], [5, 15]] is a singular matrix.

Solution

$$A = \begin{bmatrix} x & 6 \\ 5 & 15 \end{bmatrix}$$

$$|A| = \begin{vmatrix} x & 6 \\ 5 & 15 \end{vmatrix}$$

$= (x)(15) - (6)(5)$

$= 15x - 30$

A is singular, so its determinant is zero

$15x - 30 = 0$

$15x = 30$

$x = 2$

Answer: x = 2

4. Find the adjoint of the following matrices.

To find the adjoint of a 2 × 2 matrix, interchange the two diagonal elements and change the signs of the other two elements.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad B = \begin{bmatrix} 5 & -2 \\ -3 & 2 \end{bmatrix}$$

(i) A = [[a, b], [c, d]]

Interchange a and d, and change the signs of b and c

$$\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Answer: adj A = [[d, −b], [−c, a]]

(ii) B = [[5, −2], [−3, 2]]

Interchange 5 and 2, and change the signs of −2 and −3

$$\text{adj } B = \begin{bmatrix} 2 & 2 \\ 3 & 5 \end{bmatrix}$$

Answer: adj B = [[2, 2], [3, 5]]

5. Find multiplicative inverse of the following matrices.

A^{-1} = Adj A ÷ |A|. First find |A|, then adj A, and divide every element of adj A by |A|. The inverse exists only when $|A| \neq 0$.

(i) A = $\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$

studysapiens.com

$$|A| = \begin{vmatrix} 5 & 0 \\ 0 & 5 \end{vmatrix}$$

$= (5)(5) - (0)(0)$

$= 25$

$$\text{adj } A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

interchange 5 and 5 (they are equal), and $-0 = 0$

$A^{-1} = \frac{1}{25} \times \text{adj } A$

Answer: $A^{-1} = \frac{1}{25}\begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$

(ii) $\mathbf{A} = \begin{bmatrix} -4 & 8 \\ 7 & 2 \end{bmatrix}$

$$|A| = \begin{vmatrix} -4 & 8 \\ 7 & 2 \end{vmatrix}$$

$= (-4)(2) - (8)(7)$

$= -64$

$$\text{adj } A = \begin{bmatrix} 2 & -8 \\ -7 & -4 \end{bmatrix}$$

interchange -4 and 2, and change the signs of 8 and 7

$A^{-1} = \frac{-1}{64} \times \text{adj } A$

Answer: $A^{-1} = \frac{-1}{64}\begin{bmatrix} 2 & -8 \\ -7 & -4 \end{bmatrix}$

(iii) $\mathbf{A} = \begin{bmatrix} 40 & 8 \\ 5 & 2 \end{bmatrix}$

$$|A| = \begin{vmatrix} 40 & 8 \\ 5 & 2 \end{vmatrix}$$

$= (40)(2) - (8)(5)$

$= 40$

$$\text{adj } A = \begin{bmatrix} 2 & -8 \\ -5 & 40 \end{bmatrix}$$

interchange 40 and 2, and change the signs of 8 and 5

studysapiens.com

$A^{-1} = \frac{1}{40} \times \text{adj } A$

Answer: $A^{-1} = \frac{1}{40}\begin{bmatrix} 2 & -8 \\ -5 & 40 \end{bmatrix}$

(iv) $A = \begin{bmatrix} 3 & 5 \\ 5 & -3 \end{bmatrix}$

$|A| = \begin{vmatrix} 3 & 5 \\ 5 & -3 \end{vmatrix}$

$= (3)(-3) - (5)(5)$

$= -34$

$\text{adj } A = \begin{bmatrix} -3 & -5 \\ -5 & 3 \end{bmatrix}$

interchange 3 and −3, and change the signs of 5 and 5

$A^{-1} = \frac{-1}{34} \times \text{adj } A$

Answer: $A^{-1} = \frac{-1}{34}\begin{bmatrix} -3 & -5 \\ -5 & 3 \end{bmatrix}$

(v) $A = \begin{bmatrix} 10 & 8 \\ 3 & 3 \end{bmatrix}$

$|A| = \begin{vmatrix} 10 & 8 \\ 3 & 3 \end{vmatrix}$

$= (10)(3) - (8)(3)$

$= 6$

$\text{adj } A = \begin{bmatrix} 3 & -8 \\ -3 & 10 \end{bmatrix}$

interchange 10 and 3, and change the signs of 8 and 3

$A^{-1} = \frac{1}{6} \times \text{adj } A$

Answer: $A^{-1} = \frac{1}{6}\begin{bmatrix} 3 & -8 \\ -3 & 10 \end{bmatrix}$

(vi) $A = \begin{bmatrix} -2 & -3 \\ 4 & 5 \end{bmatrix}$

studysapiens.com

$$|A| = \begin{vmatrix} -2 & -3 \\ 4 & 5 \end{vmatrix}$$

$= (-2)(5) - (-3)(4)$

$= 2$

$$\text{adj } A = \begin{bmatrix} 5 & 3 \\ -4 & -2 \end{bmatrix}$$

interchange -2 and 5, and change the signs of -3 and 4

$$A^{-1} = \frac{1}{2} \times \text{adj } A$$

Answer: $A^{-1} = \frac{1}{2}\begin{bmatrix} 5 & 3 \\ -4 & -2 \end{bmatrix}$

6. If A = [[5, −3], [2, −1]], then find A^{-1} and prove that $AA^{-1} = A^{-1}A = I$.

Solution

$$A = \begin{bmatrix} 5 & -3 \\ 2 & -1 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 5 & -3 \\ 2 & -1 \end{vmatrix}$$

$= (5)(-1) - (-3)(2)$

$= -5 + 6$

$= 1$

$$\text{adj } A = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

Interchange 5 and -1, and change the signs of -3 and 2

$$A^{-1} = \frac{1}{1}\begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix}$$

Verification

$$AA^{-1} = \begin{bmatrix} 5 & -3 \\ 2 & -1 \end{bmatrix} \times \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$A^{-1}A = \begin{bmatrix} -1 & 3 \\ -2 & 5 \end{bmatrix} \times \begin{bmatrix} 5 & -3 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Answer: $AA^{-1} = A^{-1}A = I$ (proved)

studysapiens.com

7. Show that the following matrices are multiplicative inverse of each other.

Two matrices are multiplicative inverse of each other when their product is the unit matrix I (in both orders).

(i) A = [[2, −1], [−3, 2]] and B = [[2, 1], [3, 2]]

$$AB = \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$$

Row 1 × Column 1 = (2)(2) + (−1)(3) = 4 − 3 = 1

Row 1 × Column 2 = (2)(1) + (−1)(2) = 2 − 2 = 0

Row 2 × Column 1 = (−3)(2) + (2)(3) = −6 + 6 = 0

Row 2 × Column 2 = (−3)(1) + (2)(2) = −3 + 4 = 1

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$BA = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix}$$

Row 1 × Column 1 = (2)(2) + (1)(−3) = 1

Row 1 × Column 2 = (2)(−1) + (1)(2) = 0

Row 2 × Column 1 = (3)(2) + (2)(−3) = 0

Row 2 × Column 2 = (3)(−1) + (2)(2) = 1

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Answer: AB = BA = I, so A and B are multiplicative inverse of each other

(ii) A = [[4, 15], [2, 8]] and B = [[4, −15/2], [−1, 2]]

$$A = \begin{bmatrix} 4 & 15 \\ 2 & 8 \end{bmatrix} \quad B = \begin{bmatrix} 4 & -15/2 \\ -1 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 4 & 15 \\ 2 & 8 \end{bmatrix} \times \begin{bmatrix} 4 & -15/2 \\ -1 & 2 \end{bmatrix}$$

Row 1 × Column 1 = (4)(4) + (15)(−1) = 16 − 15 = 1

Row 1 × Column 2 = (4)(−15/2) + (15)(2) = −30 + 30 = 0

Row 2 × Column 1 = (2)(4) + (8)(−1) = 8 − 8 = 0

Row 2 × Column 2 = (2)(−15/2) + (8)(2) = −15 + 16 = 1

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

studysapiens.com

$$BA = \begin{bmatrix} 4 & -15/2 \\ -1 & 2 \end{bmatrix} \times \begin{bmatrix} 4 & 15 \\ 2 & 8 \end{bmatrix}$$

Row 1 × Column 1 = (4)(4) + (−15/2)(2) = 16 − 15 = 1

Row 1 × Column 2 = (4)(15) + (−15/2)(8) = 60 − 60 = 0

Row 2 × Column 1 = (−1)(4) + (2)(2) = −4 + 4 = 0

Row 2 × Column 2 = (−1)(15) + (2)(8) = −15 + 16 = 1

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Answer: AB = BA = I, so A and B are multiplicative inverse of each other

8. Prove that $(AB)^{-1} = B^{-1}A^{-1}$, if:

We find AB and its inverse, then B^{-1} and A^{-1} separately, and show that both sides come out the same.

(i) A = [[3, −2], [5, 6]] and B = [[2, −1], [−3, 2]]

$$AB = \begin{bmatrix} 3 & -2 \\ 5 & 6 \end{bmatrix} \times \begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} = \begin{bmatrix} 12 & -7 \\ -8 & 7 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} 12 & -7 \\ -8 & 7 \end{vmatrix}$$

= (12)(7) − (−7)(−8)

= 84 − 56

= 28

Adj (AB): interchange 12 and 7, and change the signs of −7 and −8

$$\text{adj } (AB) = \begin{bmatrix} 7 & 7 \\ 8 & 12 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{28} \begin{bmatrix} 7 & 7 \\ 8 & 12 \end{bmatrix}$$

Now for B: |B| = (2)(2) − (−1)(−3)

= 4 − 3

= 1

$$\text{adj } B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \quad \text{so} \quad B^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$$

For A: |A| = (3)(6) − (−2)(5)

= 18 + 10

= 28

studysapiens.com

$$\text{adj } A = \begin{bmatrix} 6 & 2 \\ -5 & 3 \end{bmatrix} \quad \text{so} \quad A^{-1} = \frac{1}{28}\begin{bmatrix} 6 & 2 \\ -5 & 3 \end{bmatrix}$$

$$B^{-1}A^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} \times \frac{1}{28}\begin{bmatrix} 6 & 2 \\ -5 & 3 \end{bmatrix}$$

Multiply the matrices first, then multiply by 1/28

$$= \frac{1}{28} \times \begin{bmatrix} 7 & 7 \\ 8 & 12 \end{bmatrix}$$

$= (AB)^{-1}$

Answer: $(AB)^{-1} = B^{-1}A^{-1}$ (proved)

(ii) A = [[1, 2], [8, 0]] and B = [[0, −1], [5, 2]]

$$AB = \begin{bmatrix} 1 & 2 \\ 8 & 0 \end{bmatrix} \times \begin{bmatrix} 0 & -1 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} 10 & 3 \\ 0 & -8 \end{bmatrix}$$

$$|AB| = \begin{vmatrix} 10 & 3 \\ 0 & -8 \end{vmatrix}$$

$= (10)(-8) - (3)(0)$

$= -80$

Adj (AB): interchange 10 and −8, and change the signs of 3 and 0

$$\text{adj } (AB) = \begin{bmatrix} -8 & -3 \\ 0 & 10 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{-80}\begin{bmatrix} -8 & -3 \\ 0 & 10 \end{bmatrix} = \frac{1}{80}\begin{bmatrix} 8 & 3 \\ 0 & -10 \end{bmatrix}$$

Now for B: $|B| = (0)(2) - (-1)(5)$

$= 0 + 5$

$= 5$

$$\text{adj } B = \begin{bmatrix} 2 & 1 \\ -5 & 0 \end{bmatrix} \quad \text{so} \quad B^{-1} = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ -5 & 0 \end{bmatrix}$$

For A: $|A| = (1)(0) - (2)(8)$

$= -16$

$$\text{adj } A = \begin{bmatrix} 0 & -2 \\ -8 & 1 \end{bmatrix} \quad \text{so} \quad A^{-1} = \frac{1}{-16}\begin{bmatrix} 0 & -2 \\ -8 & 1 \end{bmatrix} = \frac{1}{16}\begin{bmatrix} 0 & 2 \\ 8 & -1 \end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{5}\begin{bmatrix} 2 & 1 \\ -5 & 0 \end{bmatrix} \times \frac{1}{16}\begin{bmatrix} 0 & 2 \\ 8 & -1 \end{bmatrix}$$

studysapiens.com

Multiply the matrices first, then multiply the scalars

$$= \frac{1}{80} \times \begin{bmatrix} 8 & 3 \\ 0 & -10 \end{bmatrix}$$

$= (AB)^{-1}$

Answer: $(AB)^{-1} = B^{-1}A^{-1}$ (proved)

studysapiens.com